

Impacts of Public Early Childhood Education in Chile on Maternal Fertility and Labor Market Outcomes

Short-run effects — progress update

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Where we are

What we do

Question: does opening a public preschool nearby change mothers' labor supply and fertility?

Variation: 447 neighborhood units (UVs) open their *first* JUNJI/Integra preschool in 2015–2024 vs. 4,145 that never do

Design: staggered event study at the UV level, Sun–Abraham estimator (Baker et al., *JEL* 2026)

Data: woman-level panel linking administrative registries — RSH/FPS, SII/AFP earnings, Civil Registry births, preschool enrollment

What we find so far

Balance: in 2014, women in treated and never-treated UVs look alike; UVs differ in *size* → UV + year FE

First stage: enrollment rises $\approx 2\text{pp}$ on a 33% base ($\approx 6\%$) the year a preschool opens, persistent and robust across 3 estimators

Labor & fertility: estimates in progress

Project status

Done: data construction, treatment definition, balance, first stage

In progress: labor market and fertility event studies

Next: heterogeneity, mechanisms, writing

Research question

Question

What are the effects of the public expansion of early childhood education in Chile on mothers' labor market participation and fertility?

What we know

Mixed evidence: positive effects on maternal employment in Quebec and Argentina, close to zero in Norway

Recent Latin American evidence comes from **lotteries and experiments** (Rio de Janeiro, Nicaragua)

Less is known about a **national, decade-long public expansion** observed in administrative data

Our contribution

First study to use neighborhood-unit (UV) variation: preschools geolocated to the UV and linked to individual administrative records

Fertility as a second margin, alongside labor supply

What is a UV? *Unidad Vecinal*: the official sub-municipal territorial unit — ~6,900 UVs vs. 346 municipalities.

⇒ **Finer geography: a sharper measure of exposure and more variation to identify the effect.**

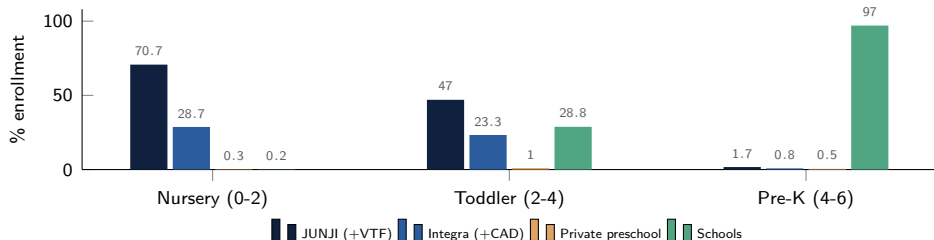
Quebec: Baker et al. (2008); Argentina: Berlinski & Galiani (2007); Norway: Havnes & Mogstad (2011); Rio: Attanasio et al. (2025); Nicaragua: Hojman & López Bóo (2022).

Public provision expanded sharply between 2015 and 2024

Mixed provision: **public** (JUNJI, Integra, municipal/SLEP) vs. **private** (subsidized/paid)

700+ new public preschools opened between 2015 and 2024

Net attendance rate (ages 0–5) rose from **50% to 55%** over the period, despite pandemic-era setbacks



Source: authors' elaboration based on Figura 5, Matrícula Oficial 2023, Centro de Estudios MINEDUC.

⇒ **Ages 0–4: JUNJI and Integra are the main providers — our focus.**

Identification: staggered-adoption event study

Design: staggered-adoption difference-in-differences with event study

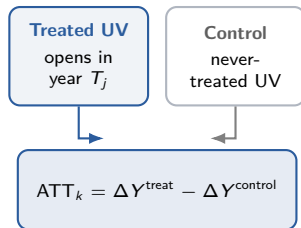
Unit of treatment: the UV, the year it opens its first public preschool

Unit of analysis: $UV \times year$ panel, built from a woman $\times$ year panel linked to each woman's UV of residence

Fixed effects: $UV + year$ (baseline); woman FE on the woman $\times$ year panel as a **robustness check**

Estimand: ATT of UV-level treatment (Baker et al., *JEL* 2026)
— an **intent-to-treat effect of access**, not individual take-up

⇒ **We compare changes in UVs that get their first preschool vs. UVs that never do — an intent-to-treat effect of access.**



A UV is treated when it opens its first preschool

Treatment definition

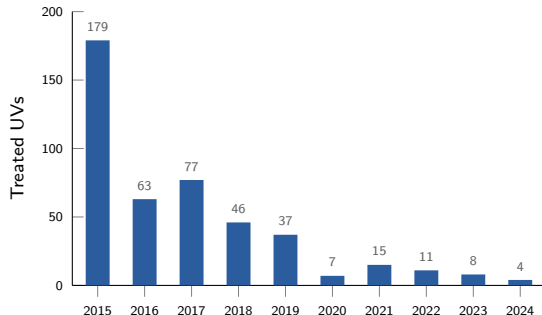
A UV is **treated** the year it opens its first public preschool (JUNJI/Integra) — goes from 0 to 1+ centers.

Group	Definition	N (UVs)	%
Excluded	Already had 1+ center in 2014 (out of design)	2,265	33.0%
Never-treated	0 in 2014, stays at 0 through 2024	4,145	60.5%
Treated	0 in 2014, opens first center 2015–2024	447	6.5%
Total universe		6,857	100%

Source: base_cobertura_cp.dta. Design sample (muestra_t1==1) = Never-treated + Treated = 4,592 UVs.

⇒ We identify the effect of the first preschool (0 → 1): 447 treated vs. 4,145 never-treated.

Treatment cohorts by year



2015 cohort is the largest: 179 UVs (40%), tied to the Bachelet preschool expansion

71% open in 2015–2017; only 45 UVs open after 2019

2014 as base year lets us identify the 2015 cohort as $0 \rightarrow 1$; without it, treated UVs drop from 447 to ~ 268

Source: base_cobertura_cp.dta. 447 treated UVs (0 centers in 2014).

⇒ **Most openings are early (2015–17): long post-treatment windows for most treated UVs.**

Empirical specification and estimator

$$Y_{ut} = \lambda_u + \lambda_t + \gamma \overline{\text{Age}}_{ut} + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - G_u = k\} + \varepsilon_{ut}$$

Y_{ut} : average outcome of women in UV u , year t (built from the woman $\times$ year panel). λ_u, λ_t : UV and year fixed effects; SE clustered by UV.

$\overline{\text{Age}}_{ut}$: average age of mothers in the UV-year (labor market specification).

β_k : effect k years from opening, relative to $k = -1$ (omitted); $k \leq -5$ binned.

Estimation strategy

Baseline TWFE as reference — but with staggered treatment, TWFE can use already-treated units as controls (Goodman-Bacon problem). Robust estimator: **Sun–Abraham** (`eventstudyinteract`), never-treated UVs as control group. Framework: Baker, Callaway, Cunningham, Goodman-Bacon & Sant’Anna (*JEL*, 2026).

Main threat: non-random placement of openings (JUNJI/Integra prioritizing UVs by care demand or socioeconomic profile). **UV FE** absorb permanent differences across UVs; **pre-period coefficients** ($k < -1$) test whether trends were already diverging.

⇒ **Main estimator: Sun–Abraham with never-treated UVs as controls; TWFE as reference.**

Data: what we have and how each sample is built

Sample	Source data	Key variables
Coverage (UV $\times$ year)	JUNJI/Integra transparency requests + UV shapefile (MDSF)	Number of centers, treatment status, opening cohort
Women panel (base)	RSH (2016–2024) + FPS (2014–2015)	UV of residence, UV population, child density
Labor market	+ Rentas (SII/AFC) + Cotizaciones (AFP)	Annual income, months worked, employment
Fertility	+ Civil Registry birth records (via RIS)	Birth in the year, children under 5
First stage (enrollment)	+ Matrícula parvularia (child-level)	Enrollment, enrollment rate

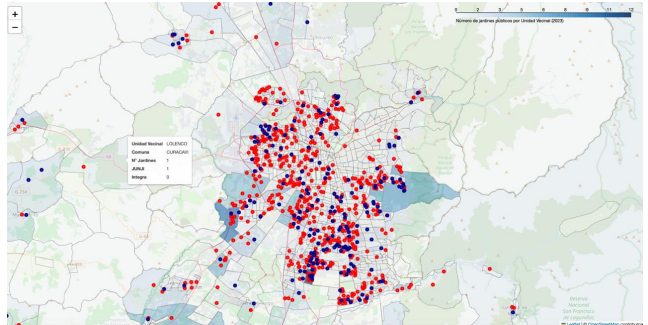
⇒ All administrative registries, linked at the individual level — no survey self-reports.

Every preschool is matched to its 2024 neighborhood unit

Each JUNJI/Integra preschool is **geolocated** and matched to the UV where it sits (spatial join)

UV boundaries **changed a lot** over time
→ we traced them and mapped everything to **2024 boundaries**

Preschools and women's UV of residence share **one common geography** across all years

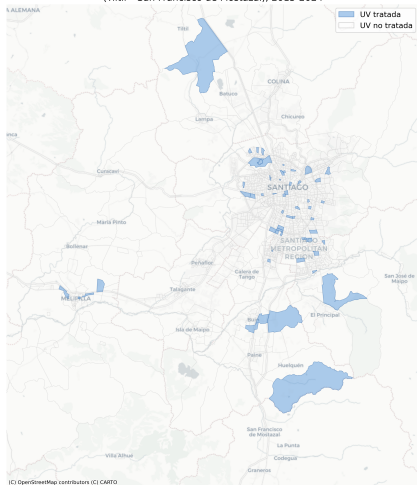


Región Metropolitana. ● JUNJI ● Integra Shading: public preschools per UV. Source: base_cobertura_cp.dta.

⇒ **One common geography: every preschool and every woman mapped to 2024 UV boundaries.**

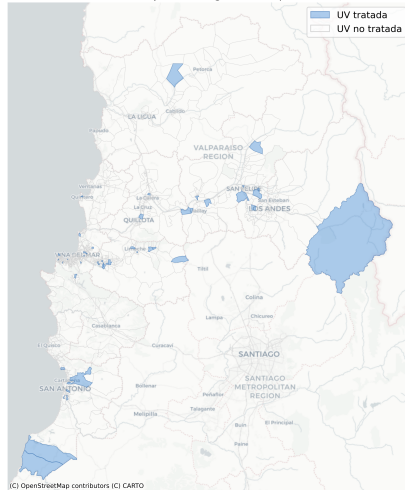
Región Metropolitana

UVs tratadas ($0 \rightarrow \geq 1$ jardín), Gran Santiago
(Tiltil - San Francisco de Mostazal), 2015-2024



Valparaíso

UVs tratadas ($0 \rightarrow \geq 1$ jardín), Región de Valparaíso, 2015-2024



Illustrative examples. Shaded: treated UVs ($0 \rightarrow 1+$ preschools), 2015–2024.

Who is in each sample, and what we measure

1. **Samples** (selected at the individual level)

Fertility: women of fertile age (18–49)

Labor market: women with a child < 5 *that specific year* — not “ever had a child”

First stage: children of treatment-sample mothers, birth to birth+4

2. **Outcomes** (averaged at $UV \times \text{year}$)

Fertility: share of women who had a child that year (Civil Registry birth records)

Labor market: average annual income, months worked, share employed, dependent vs. independent

First stage: enrollment rate

⇒ **Each outcome is measured on its relevant population, from administrative records, averaged at the $UV \times \text{year}$ level.**

Treated and never-treated UVs look alike — except in size

Labor market sample (mothers with a child <5) — covariate levels in 2014, before any opening

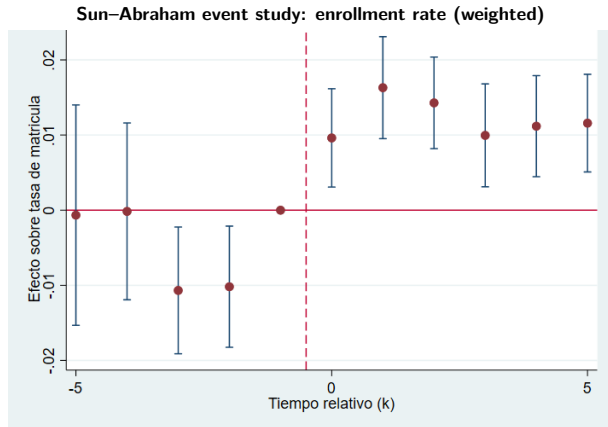
Variable	Unweighted			Weighted by UV population			Sig. ^a
	Never-tr.	Treated	Norm. diff.	Never-tr.	Treated	Norm. diff.	
Annual income (M CLP)	3.60	3.55	-0.033	3.95	3.89	-0.039	*
Months worked	8.14	8.10	-0.024	8.44	8.37	-0.058	*
Age	29.24	29.25	0.008	29.26	29.29	0.026	
N. children <5	1.11	1.11	0.045	1.11	1.11	0.049	
Hours worked	37.07	37.68	0.099	36.55	36.81	0.068	
Works (share)	0.45	0.45	0.023	0.48	0.48	-0.006	
Education (code)	4.98	5.02	0.028	5.20	5.18	-0.020	
UV population	1126.79	2011.08	0.410	3530.92	5287.26	0.340	***
Child density 0–5 UV	0.06	0.06	0.155	0.06	0.06	0.139	***
N (UVs)	3,909 / 422			3,909 / 422			

^a t-test at the woman level (* $p < 0.05$, *** $p < 0.01$), for reference only; normalized differences prioritized (concern threshold: 0.25).

The UV population gap is **expected**: the State prioritizes new preschools where need and population are larger. Same pattern in the fertility sample (App. A7).

⇒ **Selection is in *where* preschools open (UV size), not in *who* lives there → UV fixed effects.**

Enrollment rises the year a preschool opens and stays higher



Opening a preschool raises the enrollment rate of children aged 0–4 by $\approx 1\text{pp}$ vs. never-treated UVs, on a base of $\sim 33\%$.

The effect appears in the **opening year** and persists for **at least five years**.

Sun–Abraham, weighted by number of children, never-treated UVs as controls, UV and year FE, 95% CI clustered by UV.

Reference $k = -1$.

The enrollment effect is robust across three estimators

k	Unweighted TWFE	Weighted TWFE	Sun-Abraham (unweighted)
-5	-0.002	-0.015	0.006
-4	-0.009	-0.011*	0.004
-3	-0.013 [†]	-0.018***	-0.005
-2	-0.023***	-0.017***	-0.017***
0	0.020***	0.007*	0.023***
1	0.023***	0.013***	0.026***
2	0.016***	0.012***	0.019***
3	0.018***	0.007**	0.021***
4	0.017***	0.009**	0.020***
5	0.019***	0.009***	0.023***

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, [†] $p = 0.066$. Base rate $\approx 32.6\%$. $N = 48,348$ UV-year, clusters = 4,568 UVs.

Effect: $\approx 2\text{pp}$ on a $\approx 33\%$ base ($\approx 6\%$ relative), robust across specifications and growing slightly over time. Source: `primera_etapa.txt`.

The honest problem: $k = -2$ does not vanish

$k = -2$ is **negative and significant in all three specifications**, and **survives Sun–Abraham** — the estimator designed to remove cross-cohort comparison bias

By contrast, $k = -3$ loses significance under unweighted Sun–Abraham ($p=0.066 \rightarrow p=0.429$)

Hypotheses under investigation (none confirmed)

1. **Registration lag / construction transition**: the recorded opening year may lag actual full-capacity operation
2. **Age-cutoff measurement** (edad_30_06): possible source of noise, no confirmed mechanical link to $k = -2$ specifically
3. **Partial COVID overlap**: weak, covers only cohorts with $g1 \in [2021, 2023]$

Next steps

Immediate

Finish labor market estimation

Finish fertility estimation

Short term

Disaggregate $k = -2$ by opening cohort

Heterogeneity by education level (nursery/toddler vs. pre-K)

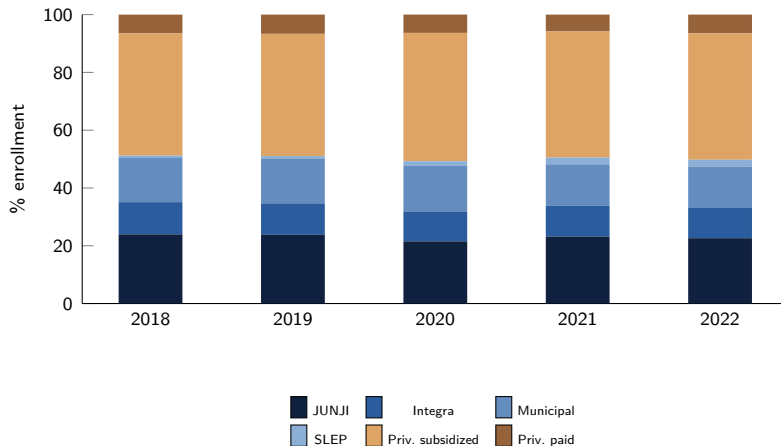
Medium term

Robustness: weighted vs. unweighted, outlier UVs

Write results section

Appendix

Appendix A1. Enrollment by dependency, 2018–2022



Source: Tabla 14, Matrícula Oficial Educación Parvularia por dependencia 2018–2022, Centro de Estudios MINEDUC.

Appendix A2. Full identification assumptions and placebo check

Parallel trends: assessed via pre-opening event-study coefficients; should be statistically indistinguishable from zero

No anticipation: mothers do not adjust decisions before the effective opening — depends on how much advance notice openings receive (to analyze)

Exogeneity of timing: opening timing should not correlate with unobserved UV characteristics. **Main threat:** JUNJI/Integra may prioritize UVs with higher care demand or specific socioeconomic profiles

Placebo check

An analogous event study will be estimated using the **pre-treatment birth rate** of the UV as the outcome. Non-significant pre-treatment coefficients would strengthen the parallel-trends assumption of the main model.

Appendix A3. Weighted vs. unweighted robustness

Weighted specification (by `n_ninos`) shows a visibly worse pre-trend: 3 of 4 pre-period coefficients significant, vs. 1 of 4 unweighted

Consistent with a small number of large outlier UVs dominating the weighted average:
`n_ninos` median 40, mean 78, p99 541, max $\approx 3,200$

Pending: robustness check excluding or capping the largest UVs

Appendix A4. Data limitation: UV assignment in 2014–2015

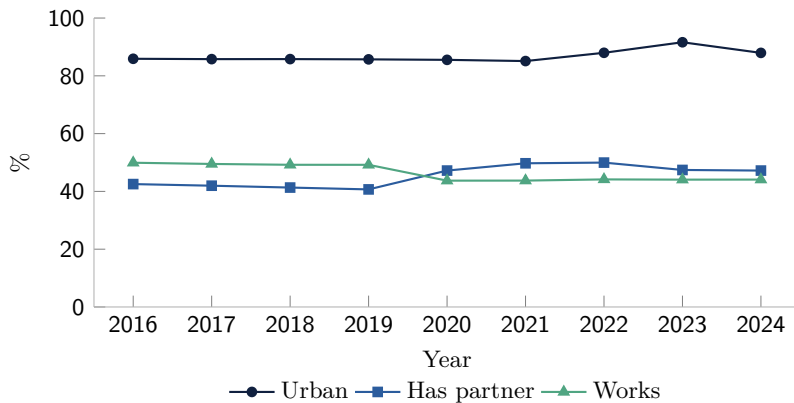
FPS (2014–2015) has no reliable UV field → those two years are anchored to the woman's **2016 UV** as a proxy

Consequence: women who left the panel before 2016 are dropped from 2014–2015 — precisely the earliest pre-trend years

RIS has not delivered a better historical crosswalk despite repeated requests; only unstandardized UV name strings are available

Idea, not yet tested: fuzzy matching on UV name strings

Appendix A5. Panel composition over time



Panel of women aged 18–70 (RSH), 2016–2024.

Appendix A6. Task status

Block	Task	Status	Note
2. Data generation	Coverage base (preschools $\times$ UV, 2014–2024)	ready	Verified; private preschools (transparency) still a possible extension
	Women panel (2014–2024, incl. FPS anchor)	ready	2014–2015 anchored to 2016 UV (see A5)
	Rentas / Cotizaciones cleaning	advanced	Independent/state/FF.AA. cotizaciones still pending
	Children registry cleaning	ready	tuvo_hijo, n_hijos_menor5
	Merge panel + outcome variables	ready	By rut_inn
	Treated / non-treated definition	ready	grupo_t1, g1, muestra_t1
3. Panel analysis	First-stage estimation (enrollment)	ready	3 specifications; $k = -2$ open issue
	Balance / validity check	ready	UV-level balance table
4. Data analysis	Methodology choice	ready	Sun–Abraham within Baker et al. (JEL 2026) framework
	Treatment definition	ready	Extensive margin, $0 \rightarrow 1$
	Run regressions (labor, fertility)	in progress	Expected before Thursday
	Analyze results	in progress	First stage analyzed; rest pending
5. Mechanisms	Heterogeneity by group / level	not started	
	Public-policy implications	not started	
6. Writing	—	in progress	

Appendix A7. Balance in the fertility sample

Fertility sample (women aged 18–49) — covariate levels in 2014, before any opening

Variable	Unweighted			Weighted by UV population			Sig. ^a
	Never-tr.	Treated	Norm. diff.	Never-tr.	Treated	Norm. diff.	
Birth rate	0.057	0.058	0.031	0.06	0.060	0.051	
Age	31.90	31.83	-0.053	31.74	31.82	0.095	
Children <5 per woman	0.303	0.319	0.156	0.31	0.320	0.117	**
UV population	1111.42	2025.96	0.423	3540.63	5305.76	0.342	***
Child density 0–5 UV	0.0590	0.0631	0.176	0.06	0.064	0.148	**
Women in sample per UV	233.60	445.45	0.401	795.29	1234.46	0.347	***
N (UVs)	3,973 / 422			3,973 / 422			

^a t-test at the UV level (** $p < 0.01$, *** $p < 0.001$), for reference only; normalized differences prioritized (concern threshold: 0.25). UVs with at least one sample woman in 2014, hence more never-treated UVs than in the labor sample.

⇒ **Same conclusion as the labor sample: women look alike, UVs differ in size.**